

Document filename: 290426ab.sqx
Spectrum ID:
Model ID: Default Model

EXPERIMENTAL CONDITIONS			
Excitation mode: X-ray fluorescence (XRF)			
Sample detector path: Air		Detector type: Si(Li)	
Path length: 3.0 cm		Dead layer: 0.0300 μm	
Primary energy: 16.5 Kev		Contact layer: 0.0200 μm	
Incidence angle: 45.0 degrees		Active depth: 3.0 mm	
Take-off angle: 45.0 degrees		Window: 760.00 μm	
Pulse pile-up resolution time: 2.00 μs			

Fitting Region: channels 91 - 1634, 4 iterations done, last change = 0.04%

Lambda = 1e-006 , ChiSquare = 4.4

CALIBRATION DATA			
	Initial estimate	Final estimate	
ZERO (eV)	-387.2 ± 100.0	-390.6 ± 0.3	
GAIN (eV/ch)	14.673 ± 2.000	14.648 ± 0.001	
NOISE (eV)	81.5 ± 40.0	50.7 ± 2.9	
FANO factor	0.155 ± 0.050	0.148 ± 0.003	

PEAK DATA					
#	Line	E(KeV) chan#	Rel. int. fwhm (eV)	Peak area st. dev backgr	Chi-sq tot. abs

1	P K			623 ± 65	
	KA1	2.013	1.00000	623 ± 65	1.4
		164.092	94.50	1501	1.49e-005

2	S K			10502 ± 172	
	KA1	2.307	0.80932	8499 ± 113	21.9
		184.163	99.29	2379	5.81e-004
	KB1	2.464	0.19068	2002 ± 59	72.1
		194.882	101.75	2921	2.22e-003

3	Cl K			21476 ± 249	
	KA1	2.622	0.82302	17675 ± 155	15.9
		205.668	104.18	3963	6.32e-003
	KB1	2.816	0.17464	3751 ± 73	51.7
		218.912	107.08	4402	1.66e-002
	KB1 Escape	1.074	0.00235	50 ± 21	1.7
		99.988	77.23	438	7.09e-027

4	Ar K			14349 ± 374	
	KA1	2.957	0.81836	11742 ± 245	21.6
		228.538	109.14	4574	2.91e-002
	KB1	3.191	0.16920	2428 ± 83	33.0
		244.513	112.47	4538	5.90e-002
	KA1 Escape	1.215	0.01046	150 ± 22	3.9
		109.614	80.06	467	1.92e-022
	KB1 Escape	1.449	0.00199	28 ± 24	2.6
		125.589	84.55	573	1.45e-013

5	K K			85678 ± 455	
	KA1	3.313	0.82516	70698 ± 306	16.9
		252.842	114.17	4398	7.89e-002
	KB1	3.590	0.16393	14045 ± 86	8.2
		271.752	117.94	3806	1.36e-001
	KA1 Escape	1.571	0.00925	793 ± 28	7.2
		133.918	86.80	787	8.15e-011
	KB1 Escape	1.848	0.00165	142 ± 34	1.0
		152.828	91.70	1168	5.92e-007

6	Ca K			206862 ± 586	
	KA1	3.691	0.83279	172272 ± 408	8.9
		278.647	119.28	3522	1.59e-001
	KB1	4.013	0.15777	32637 ± 94	22.1
		300.630	123.47	2986	2.40e-001
	KA1 Escape	1.949	0.00809	1673 ± 37	1.1
		159.723	93.42	1365	4.84e-006
	KB1 Escape	2.271	0.00135	280 ± 47	21.0
		181.706	98.71	2205	4.18e-004

7	Ti K			536 ± 253	
	KA1	4.509	0.85258	457 ± 68	1.1
		334.491	129.66	2307	3.66e-001
	KB1	4.932	0.14056	75 ± 51	0.7
		363.369	134.71	2572	4.65e-001
	KA1 Escape	2.767	0.00602	3 ± 66	49.9
		215.567	106.35	4322	1.33e-002

KB1 Escape	3.190	0.00084	0	± 67	33.0
	244.445	112.46	4538		5.89e-002
8 Cr K			342	± 251	
KA1	5.412	0.86647	297	± 79	1.5
	396.138	140.23	3304		5.60e-001
KB1	5.947	0.12866	44	± 62	1.4
	432.662	146.13	3828		6.47e-001
KA1 Escape	3.670	0.00434	1	± 60	8.9
	277.214	119.01	3563		1.54e-001
KB1 Escape	4.205	0.00053	0	± 49	3.2
	313.737	125.90	2432		2.89e-001
9 Mn K			3965	± 271	
KA1	5.895	0.86610	3434	± 104	1.4
	429.112	145.56	3784		6.39e-001
KB1	6.491	0.12983	515	± 67	2.7
	469.800	151.89	4324		7.15e-001
KA1 Escape	4.153	0.00363	14	± 51	10.1
	310.188	125.25	2554		2.76e-001
KB1 Escape	4.749	0.00043	2	± 50	1.7
	350.876	132.55	2466		4.24e-001
10Fe K			8753	± 307	
KA1	6.399	0.86901	7607	± 127	2.5
	463.519	150.93	4250		7.05e-001
KB1	7.059	0.12764	1117	± 74	3.4
	508.577	157.69	5272		7.71e-001
KA1 Escape	4.657	0.00300	26	± 49	1.1
	344.595	131.45	2409		4.02e-001
KB1 Escape	5.317	0.00035	3	± 57	1.8
	389.652	139.15	3206		5.43e-001
11 Co K			1004	± 292	
KA1	6.925	0.87101	874	± 102	3.3
	499.429	156.34	5174		7.59e-001
KB1	7.649	0.12623	127	± 76	2.6
	548.855	163.49	5634		8.15e-001
KA1 Escape	5.183	0.00248	2	± 52	1.3
	380.504	137.62	2733		5.17e-001
KB1 Escape	5.907	0.00028	0	± 62	1.4
	429.931	145.69	3784		6.41e-001
12Ni K			1239	± 432	
KA1	7.478	0.57714	715	± 90	1.8
	537.181	161.83	5546		8.04e-001
KA2	7.461	0.29531	366	± 79	2.6
	536.021	161.66	5537		8.02e-001
KB1	8.265	0.12527	155	± 78	2.4
	590.909	169.33	5901		8.51e-001
KA1 Escape	5.736	0.00135	2	± 60	0.8
	418.257	143.83	3618		6.15e-001
KA2 Escape	5.719	0.00070	1	± 60	0.8
	417.096	143.64	3603		6.12e-001
KB1 Escape	6.523	0.00023	0	± 66	2.9
	471.985	152.22	4348		7.19e-001
13Cu K			753	± 456	
KA1	8.048	0.57872	436	± 91	1.3
	576.094	167.30	5825		8.39e-001
KA2	8.028	0.29671	223	± 80	1.3
	574.729	167.11	5813		8.38e-001
KB1	8.905	0.12269	92	± 82	1.4
	634.601	175.20	6678		8.79e-001
KA1 Escape	6.306	0.00112	1	± 65	4.8
	457.170	149.96	4173		6.94e-001
KA2 Escape	6.286	0.00058	0	± 64	4.7
	455.805	149.75	4147		6.92e-001
KB1 Escape	7.163	0.00018	0	± 73	6.6
	515.677	158.72	5344		7.80e-001
14Zn K			1112	± 485	
KA1	8.639	0.57753	642	± 97	1.3
	616.441	172.79	6614		8.68e-001
KA2	8.616	0.29663	330	± 86	1.4
	614.871	172.57	6606		8.67e-001
KB1	9.572	0.12429	138	± 83	2.1
	680.136	181.11	6770		9.01e-001
KA1 Escape	6.897	0.00093	1	± 72	3.3
	497.517	156.06	5152		7.57e-001
KA2 Escape	6.874	0.00048	1	± 72	4.0
	495.947	155.82	5129		7.54e-001
KB1 Escape	7.830	0.00015	0	± 76	3.7
	561.212	165.23	5729		8.27e-001
15Se K			646	± 601	
KA1	11.222	0.56839	367	± 100	2.5
	792.780	194.97	7151		9.38e-001
KA2	11.181	0.29297	189	± 89	2.3
	789.981	194.63	7165		9.37e-001
KB1	12.494	0.13072	84	± 82	2.1
	879.617	205.01	6572		9.48e-001
KB2	12.652	0.00718	5	± 80	1.9

	890.404	206.22	6479		9.50e-001
KA1 Escape	9.480	0.00044	0	± 82	2.0
	673.855	180.31	6763		8.99e-001
KA2 Escape	9.439	0.00023	0	± 82	2.1
	671.056	179.95	6761		8.97e-001
KB1 Escape	10.752	0.00008	0	± 85	2.3
	760.693	191.12	7284		9.29e-001
16Br K			28253	± 658	
KA1	11.924	0.56311	15909	± 142	3.0
	840.704	200.57	6866		9.41e-001
KA2	11.878	0.29362	8295	± 102	2.7
	837.564	200.21	6886		9.47e-001
KB1	13.289	0.13261	3747	± 85	1.6
	933.891	211.04	6575		9.57e-001
KB2	13.470	0.01003	283	± 80	1.0
	946.248	212.39	6430		9.58e-001
KA1 Escape	10.182	0.00037	10	± 82	2.4
	721.780	186.35	6763		9.17e-001
KA2 Escape	10.136	0.00020	6	± 82	2.4
	718.640	185.96	6767		9.16e-001
KB1 Escape	11.547	0.00006	2	± 84	4.3
	814.967	197.58	7033		9.43e-001
17Sr K			1685	± 559	
KA1	14.165	0.55730	939	± 94	0.7
	993.695	217.50	5894		9.63e-001
KA2	14.098	0.28994	489	± 82	0.8
	989.121	217.01	5939		9.63e-001
KB1	15.832	0.13506	228	± 73	1.3
	1107.499	229.28	5107		9.71e-001
KB2	16.085	0.01732	29	± 70	1.5
	1124.771	231.02	4923		9.71e-001
KA1 Escape	12.423	0.00022	0	± 81	2.2
	874.770	204.46	6613		9.48e-001
KA2 Escape	12.356	0.00012	0	± 82	2.9
	870.196	203.94	6645		9.47e-001
KB1 Escape	14.090	0.00004	0	± 77	0.8
	988.574	216.95	5950		9.62e-001
18Ag K			622	± 211	
KA1	22.163	0.65194	405	± 56	0.9
	1539.708		269.38		1935
KA2	21.990	0.34795	216	± 49	1.5
	1527.898		268.36		2036
KA1 Escape	20.421	0.00007	0	± 54	10.1
	1420.784		258.97		2897
KA2 Escape	20.248	0.00004	0	± 53	5.2
	1408.974		257.91		2796
19Ag L			38260	± 1192	
L1M3	3.234	0.11071	4236	± 85	24.6
	247.449	113.07	4494		6.51e-002
L1M2	3.203	0.06168	2360	± 73	29.5
	245.332	112.64	4524		6.01e-002
L1N3	3.747	0.08968	3431	± 71	7.8
	282.470	120.02	3354		1.73e-001
L1M5	3.437	0.00275	105	± 65	4.8
	261.307	115.87	4160		1.02e-001
L2M4	3.151	0.23201	8877	± 128	50.2
	241.782	111.91	4571		5.29e-002
L2N4	3.520	0.05488	2100	± 68	4.1
	266.973	117.00	3993		1.20e-001
L2M1	2.806	0.00206	79	± 66	51.7
	218.229	106.93	4402		1.59e-002
L2N1	3.428	0.00251	96	± 65	4.8
	260.693	115.75	4191		1.01e-001
L3M5	2.984	0.33280	12733	± 170	27.3
	230.381	109.53	4590		3.21e-002
L3N5	3.348	0.09173	3510	± 79	4.9
	255.231	114.65	4328		8.54e-002
L3M1	2.634	0.00260	99	± 63	14.8
	206.487	104.36	4006		6.77e-003
L3N1	3.256	0.00516	198	± 67	24.7
	248.950	113.38	4477		6.88e-002
L1M3 Escape	1.492	0.00128	49	± 25	3.0
	128.524	85.35	605		1.71e-012
L1M2 Escape	1.461	0.00073	28	± 24	3.1
	126.408	84.77	584		2.97e-013
L1N3 Escape	2.005	0.00085	33	± 38	1.4
	163.546	94.36	1474		1.30e-005
L2M4 Escape	1.409	0.00276	106	± 23	2.3
	122.858	83.80	539		1.10e-014
L2N4 Escape	1.778	0.00057	22	± 32	1.3
	148.049	90.48	1049		1.06e-007
L3M5 Escape	1.242	0.00421	161	± 22	5.9
	111.457	80.59	473		4.60e-021
L3N5 Escape	1.606	0.00102	39	± 29	9.4
	136.307	87.43	820		3.57e-010

8.79e-001

8.85e-001

9.26e-001

9.30e-001